

NIBESH SHRESTHA

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(Last Updated: March 2026)

RESEARCH INTERESTS

My research focuses on designing efficient and secure distributed computing primitives for coordination in adversarial environments. I develop communication- and latency-efficient protocols for Byzantine fault-tolerant consensus, distributed randomness beacons, and distributed key generation, which serve as foundational building blocks for decentralized infrastructure. By combining techniques from distributed systems and cryptography, my work aims to improve the scalability, fairness, and resilience of large-scale distributed platforms such as blockchains and cloud-scale services.

EDUCATION

Rochester Institute of Technology

Ph.D., Computer Science

Advisors: Kartik Nayak (Duke), Pengcheng Shi (RIT)

Thesis Title: Efficient Synchronous Byzantine Consensus

GPA: 3.89 out of 4

Rochester, NY

August 2023

Institute of Engineering, Tribhuvan University

B.E. Electronics and Communication Engineering

GPA: 3.81 out of 4

Lalitpur, Nepal

2013

PROFESSIONAL POSITIONS

Supra Research, Applied Researcher

Duke University, Research Associate

Chainlink Labs, Research Intern

Duke University, Research Associate

Duke University, Research Associate

Upwork Global Inc., Freelance Software Engineer

Rochester, NY | August 2023 - Present

Remote | April 2023 - July 2023

Remote | August 2022 - November 2022

Remote | June 2021 - July 2021

Remote | May 2020 - July 2020

Remote | 2013 - June 2017

PUBLICATIONS

default ordering – alphabetical

Otherwise, by contribution order. (* denotes equal contribution)

Peer-Reviewed Conferences and Journals

1. Hydrangea: Optimistic Two-Round Partial Synchrony with Improved Fault Resilience

Nibesh Shrestha, Aniket Kate, Kartik Nayak

USENIX Security Symposium (USENIX Security), 2026

2. Towards Improving Throughput and Scalability of DAG-based BFT

Nibesh Shrestha, Aniket Kate

European Conference on Computer Systems (EuroSys), 2026

3. The Pipes Model for Latency Analysis

Andrew Lewis-Pye, Kartik Nayak, Nibesh Shrestha

IEEE Symposium on Security and Privacy (S&P), 2026

4. Low-Latency Dynamically Available Total Order Broadcast

Sravya Yandamuri, Nibesh Shrestha, Luca Zanolini, Kartik Nayak

Financial Cryptography (FC), 2026

5. Optimistic Signature-Free Reliable Broadcast and its Applications

Nibesh Shrestha*, Qianyu Yu*, Aniket Kate, Giuliano Losa, Kartik Nayak, Xuechao Wang

ACM Conference on Computer and Communications Security (CCS), 2025 | **Distinguished Paper Award**

6. Sailfish: Towards Improving the Latency of DAG-based BFT

Nibesh Shrestha, Rohan Shrothrium, Aniket Kate, Kartik Nayak

IEEE Symposium on Security and Privacy (S&P), 2025

7. **Communication and Round Efficient Parallel Broadcast Protocols**

Nibesh Shrestha, Ittai Abraham, Kartik Nayak
Financial Cryptography (FC), 2025

8. **Moonshot: Optimizing Block Period and Commit Latency in Chain-Based Rotating Leader BFT**

Isaac Doidge, Raghavendra Ramesh, **Nibesh Shrestha**, Joshua Tobkin
IEEE Dependable Systems and Networks (DSN), 2024

9. **Synchronous Distributed Key Generation without Broadcasts**

Nibesh Shrestha, Adithya Bhat, Aniket Kate, Kartik Nayak
IACR Communications in Cryptology, 2024

10. **OptRand: Optimistically Responsive Distributed Random Beacons**

Adithya Bhat*, **Nibesh Shrestha***, Aniket Kate, Kartik Nayak
Network and Distributed System Security Symposium (NDSS), 2023

11. **Optimal Good-case Latency for Rotating Leader Synchronous BFT**

Ittai Abraham, Kartik Nayak, **Nibesh Shrestha**
Principles of Distributed Systems (OPODIS), 2021 | **Best Paper Award**

12. **Making synchronous BFT protocols secure in the presence of mobile sluggish faults**

Justin Kim, Vandan Mehta, Kartik Nayak, **Nibesh Shrestha**
Brief announcement, Principles of Distributed Computing (PODC), 2021

13. **RandPiper: Reconfiguration-Friendly Random Beacons with Quadratic Communication**

Adithya Bhat*, **Nibesh Shrestha***, Aniket Kate, Kartik Nayak
ACM Conference on Computer and Communications Security (CCS), 2021

14. **On the Optimality of Optimistic Responsiveness**

Nibesh Shrestha, Ittai Abraham, Ling Ren, Kartik Nayak
ACM Conference on Computer and Communications Security (CCS), 2020

Preprints and Submissions

- **Efficient Distributed Key Generation for Blockchains**

Aniket Kate, Pratyay Mukherjee, Pratik Sarkar, Hamza Saleem, **Nibesh Shrestha**, David Yang

Tech Reports

1. **Revisiting hBFT: Speculative Byzantine Fault Tolerance with Minimum Cost**

Nibesh Shrestha, Mohan Kumar, SiSi Duan

2. **Hydrangea++: Enhancing Hydrangea with Optimistic Proposals**

Nibesh Shrestha, Aniket Kate

TEACHING EXPERIENCE

- Graduate Teaching Assistant

Rochester Institute of Technology | 2019 - 2023

Assisted instruction for graduate and undergraduate courses in *Analysis of Algorithms*. Led discussion sessions and graded coursework.

SOFTWARE ARTIFACTS

- Hydrangea (Rust) – <https://github.com/nibeshrestha/hydrangea>
- Sailfish (Rust) – <https://github.com/nibeshrestha/sailfish>
- OptRand (C++) – <https://github.com/nibeshrestha/optrand>
- SimpleSync BFT – <https://github.com/nibeshrestha/simplesync>
- OptSync Protocol – <https://github.com/nibeshrestha/optsync>

PROFESSIONAL SERVICE

Program Committee: FC (2025)
External Reviewer: ACM CCS (2023, 2022, 2021), IEEE S&P (2022, 2025), Eurocrypt (2025),
FC (2022, 2021), PerCom (2020), JPDC (2020)

AWARDS

- Distinguished Paper Award, ACM CCS 2025
- Best Paper Award, OPODIS 2021
- Research and Creativity Award, RIT 2022
- RIT PhD Merit Scholarship (2017–2019)
- College Fellowship Scholarship (Undergraduate)

SELECTED TALKS

1. **Sailfish: Towards Improving Latency of DAG-based BFT**
IEEE Symposium on Security and Privacy (S&P) May 2025
2. **OptRand: Optimistically Responsive Distributed Randomness**
Network and Distributed Systems Security (NDSS) February 2023
3. **Synchronous Distributed Key Generation without Broadcasts**
Crypto Economics Security Conference (CESC) 2022
4. **Optimal Good-case Latency for Rotating Leader Synchronous BFT**
Principles of Distributed Systems (OPODIS) December 2021
5. **RandPiper–Reconfiguration-Friendly Random Beacons with Quadratic Communication**
ACM Computer and Communications Security (CCS) November 2021
6. **On the Optimality of Optimistic Responsiveness**
ACM Computer and Communications Security (CCS) Novemeber 2020
Workshop on Foundations of Computer Security June 2020

References

1. Kartik Nayak  Associate Professor, Duke University
2. Aniket Kate  Associate Professor, Purdue University
3. Ittai Abraham  Senior Researcher, a16z Crypto
4. Pengcheng Shi  Professor, Rochester Institute of Technology